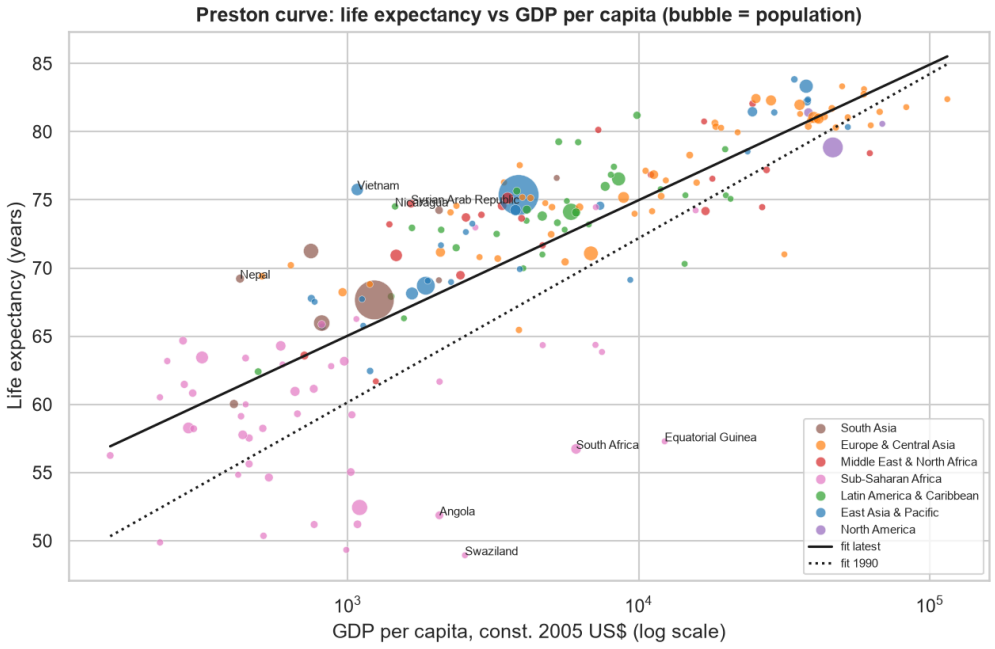
Data Analytics · Group Project G2

Does richer mean more developed?

A data-mining study of the World Development Indicators, 1960–2014



Group G2

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Data: World Bank WDI via Kaggle (`kaggle/world-development-indicators`) [1, 2].
Code, notebook and interactive app: this repository (`WDI_Analysis.ipynb`, `wdi/`, `app.py`).

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1 Introduction

The World Development Indicators (WDI) are the World Bank's main collection of country statistics [1]. We use the Kaggle extract, which covers 247 economies and 1,344 indicators from 1960 to 2015. The project asks one question:

"Does richer mean more developed?"

We answer it in three steps that follow the levels of analytics, stating the answer first [17]. **What** happened (descriptive: trends and convergence), **why** (diagnostic: how much income explains, and who does better or worse than their income predicts), and **so what** (prescriptive: data-driven country groups and Vietnam's position). Vietnam is our running case study and is drawn in **red** in every figure.

Short answer. Mostly yes. Log GDP per capita alone explains 51–77 % of the cross-country variance of the other themes (cross-validated R^2). But the remainder is where the story is: Vietnam lives 10.4 years longer than its income predicts, the highest residual of 203 countries.

Tools. Python (pandas [15], NumPy, SciPy, scikit-learn [14], XGBoost [12], LightGBM [13]), Matplotlib/Seaborn for the report figures, Plotly and Streamlit [16] for the interactive web app (Section 6).

The report follows the assignment structure: Section 2 explores the raw data (A1–A3), Section 3 lists the preprocessing techniques we applied (B1–B5), Section 4 describes the data-mining actions, Section 5 evaluates them, Section 6 presents the web app, and Sections 7–9 discuss findings, limitations and conclusions.

2 Initial data exploration

2.1 A1. Dataset overview: size and attributes

The Kaggle dataset has six CSV files and an equivalent SQLite database. We use three of them (Table 1).

Table 1: Raw files used. `database.sqlite` (1.4 GB) duplicates the CSVs and was not used.

File	Size	Rows	Columns	Content
`Indicators.csv`	574 MB	5,656,458	6	one observation per country × indicator × year
`Country.csv`	0.13 MB	247	31	country metadata (region, income group, ...)
`Series.csv`	3.3 MB	1,345	20	indicator metadata (topic, definition, unit, source)
`Footnotes, CountryNotes, SeriesNotes`	56 MB	–	–	free-text notes (not used)

Attributes. `Indicators.csv` is a long table with six attributes: `CountryName` and `CountryCode` (nominal, 247 values), `IndicatorName` and `IndicatorCode` (nominal, 1,344 values), `Year` (discrete numeric, 1960–2015) and `Value` (continuous numeric, the measurement). The meaning and unit of `Value` depend on the indicator. From `Country.csv` we use `ShortName`, `Region` (7 regions, nominal) and `IncomeGroup` (ordinal: low, lower-middle, upper-middle, high OECD/non-OECD). From `Series.csv` we use `Topic` and `IndicatorName`.

Structure. The 247 "countries" contain 214 real economies and 33 aggregates (World, Euro area, income and regional groups); aggregates have no `Region`. Coverage grows strongly over time: 23,188 observations in 1960, 117,634 in 1990, 187,206 in 2010, and only 14,762 in 2015 (the last year is mostly empty). The 1,344 indicators are dominated by "Economic Policy & Debt" (506 series); health, education and environment have about 100–120 each.

Selected attributes. Working with 1,344 sparse indicators is neither feasible nor meaningful, so we selected 32 indicators in six themes (Table 19 in the appendix): Economy (8), Demography (7),

Health (5), Education (5), Environment (4) and Technology (3). Selection criteria were relevance to development, coverage, and comparability across countries.

A1. Summary

- 5.66 M observations in long format (country, indicator, year, value), 247 economies (214 countries + 33 aggregates), 1,344 indicators, 1960–2015; 574 MB on disk.
- Attributes: two nominal keys (country, indicator), one discrete time attribute (year), one continuous value; plus nominal `Region` and ordinal `IncomeGroup` from the metadata.
- The table is very sparse and gets denser over time. We keep 32 indicators in six themes (247k rows), which become the numeric attributes of a country-year panel (11,935 × 32) and a country snapshot (214 × 32).

2.2 A2. Summary statistics of the attributes

Table 2 gives the summary statistics of the 32 attributes in the cross-country snapshot (latest value per country in 2005–2014, so each country counts once). Figure 1 shows the distributions of key attributes.

Table 2: Summary statistics of the snapshot (214 countries, latest value 2005–2014). Miss. % = share of countries with no value in the window.

Indicator	Theme	n	Miss. %	Mean	Median	Std	Min	Max	Skew
gdp_pc	Economy	199	7.0	13,104	4,112	20,883	153	158,603	3.16
gdp_pc_ppp	Economy	192	10.3	18,337	11,467	20,870	567	134,182	2.53
gdp_growth	Economy	199	7.0	3.15	3.10	3.53	-24.0	10.3	-2.43
inflation	Economy	176	17.8	4.19	2.54	7.47	-1.51	62.2	4.90
unemployment	Economy	174	18.7	8.58	6.90	6.14	0.300	31.0	1.50
trade_open	Economy	189	11.7	94.7	84.3	53.5	19.1	439	2.91
gini	Economy	145	32.2	38.6	37.4	8.65	16.6	63.4	0.50
poverty_190	Economy	118	44.9	19.2	7.83	23.1	0	81.8	1.10
population	Demography	214	0.0	33,819,246	6,183,345	133,126,384	9,893	1,364,270,000	8.86
pop_growth	Demography	214	0.0	1.31	1.25	1.24	-4.19	8.09	0.49
urban	Demography	212	0.9	58.8	58.9	24.2	8.55	100	-0.09
fertility	Demography	203	5.1	2.79	2.28	1.40	1.12	7.62	1.07
age65	Demography	195	8.9	8.23	5.88	5.74	1.04	25.7	0.87
age0_14	Demography	195	8.9	28.1	27.5	10.6	11.9	50.4	0.30
female_lfp	Demography	186	13.1	52.8	53.5	16.0	13.6	88.0	-0.26
life_exp	Health	203	5.1	71.2	73.6	8.50	48.9	83.8	-0.73
infant_mort	Health	192	10.3	24.7	15.2	22.6	1.60	98.8	1.16
u5_mort	Health	192	10.3	33.0	17.8	33.8	2.00	162	1.44
health_exp	Health	188	12.1	6.79	6.49	2.86	1.27	19.7	1.00
safe_water	Health	199	7.0	88.5	95.6	15.2	31.7	100	-1.51
enrol_primary	Education	188	12.1	105	104	14.8	29.2	165	0.16
enrol_secondary	Education	185	13.6	81.7	88.3	28.8	7.35	166	-0.37
enrol_tertiary	Education	169	21.0	37.4	33.1	27.4	0.076	117	0.47
literacy	Education	155	27.6	82.0	92.0	20.6	15.5	100.0	-1.29
edu_exp	Education	165	22.9	4.74	4.55	2.01	1.10	13.0	1.07
co2_pc	Environment	199	7.0	4.87	2.63	6.33	0.021	44.0	2.82
energy_pc	Environment	168	21.5	2,209	1,275	2,819	9.72	18,813	3.08
forest	Environment	207	3.3	32.0	32.7	24.0	0	98.3	0.48
renewables	Environment	186	13.1	32.4	22.6	29.7	0.009	96.6	0.76
electricity	Technology	212	0.9	78.2	97.4	29.8	5.06	100	-1.11
internet	Technology	204	4.7	45.2	44.7	29.4	0	98.2	0.09
mobile	Technology	205	4.2	105	106	43.2	6.39	323	0.66

Observations.

- **Strong right skew** in income, population, CO_2, energy, trade and mortality (skew 1.4–8.9). The mean GDP per capita (US$13,104) is three times the median (US$4,112). We therefore

report medians and use $\log_{10}$ for these attributes in correlation, regression and clustering (B3).

- **Bounded percentages** (electricity, water, literacy) are left-skewed with a ceiling at 100 %; enrolment ratios can exceed 100 % because they are gross ratios (over-age pupils).
- **Scales differ by orders of magnitude** (population up to 1.36×10^9, fertility 1.1–7.6), so attributes must be standardised before distance-based methods.

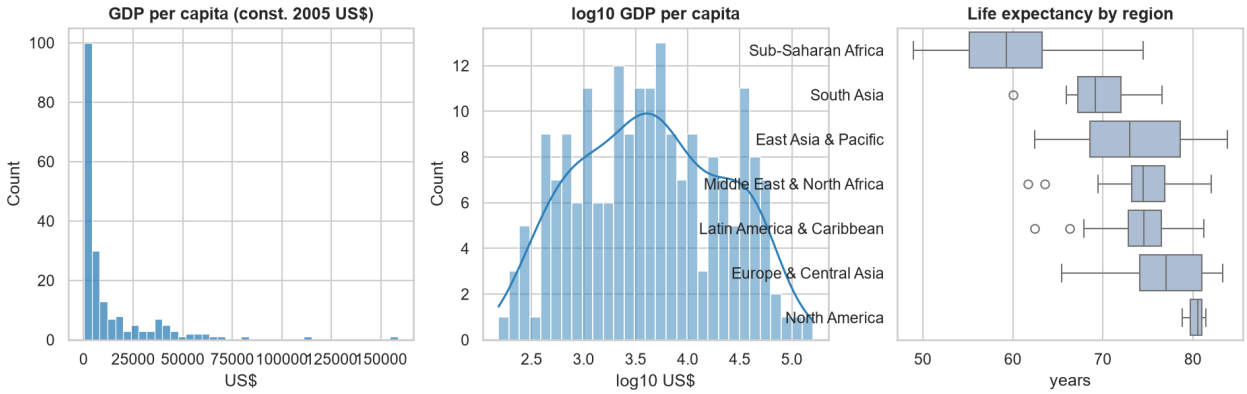


Figure 1: Distributions of key attributes in the snapshot: skewed income and CO_2 become near-symmetric on a log scale.

Missing values and their impact on the summary statistics. Missing values are the main data-quality issue. In the country-year panel, 44.9 % of the cells are empty. Most of this is *structural*: some series only start in 1990 (internet, electricity, renewables), and others come from irregular household surveys (poverty, Gini, literacy). Figure 2 shows coverage by indicator and decade.

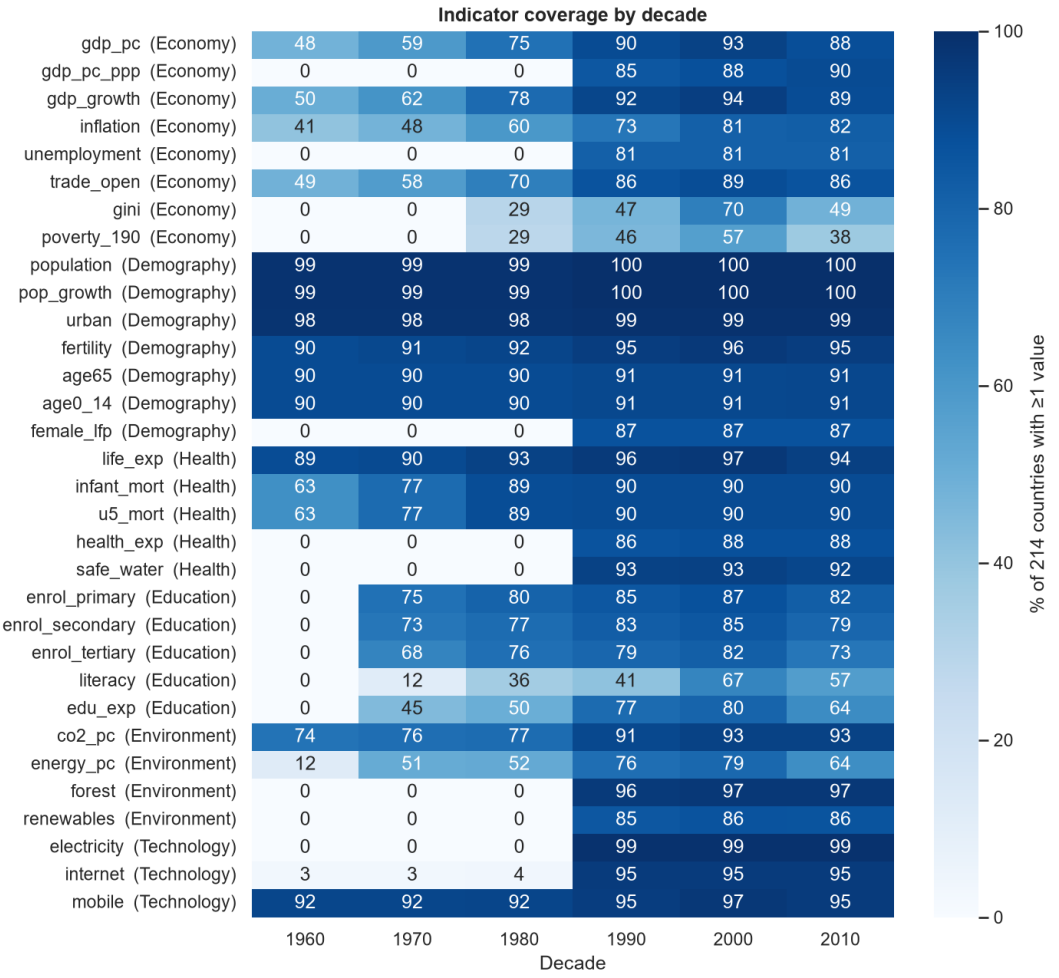


Figure 2: Share of countries reporting each indicator by decade. Survey-based indicators (bottom) are sparse in every decade.

Missingness is **not random** [6]: it depends on the income group (Figure 3). Poverty at \$1.90 is reported by 74–86 % of low and lower-middle income countries but only 19 % of high-income countries (the World Bank does not measure extreme poverty there). Literacy is reported by 44 % of high-income countries; child mortality by 75 %.

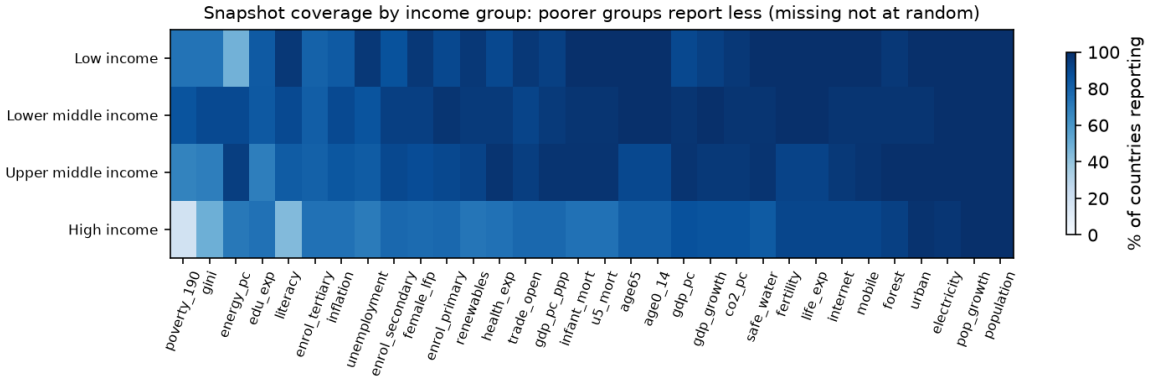


Figure 3: Snapshot coverage by World Bank income group. Darker = more countries report. Coverage depends on income, so the reporting sample is biased.

To **demonstrate the impact** on the summary statistics, Table 3 compares the mean and median computed on reporting countries only with the same statistics after filling each missing value with

the median of its income group.

Table 3: Impact of missing values on summary statistics. "rep." = reporting countries only; "imp." = after income-group median imputation; Cov. = % of countries reporting in each income group (LI, LMI, UMI, HI).

Indicator	Miss. %	Mean (rep.)	Mean (imp.)	Shift %	Median (rep.)	Median (imp.)	Cov. LI	Cov. LMI	Cov. UMI	Cov. HI
gdp_pc	7.0	13,104	13,459	+2.7	4,112	4,661	90	98	98	87
life_exp	5.1	71.2	71.6	+0.5	73.6	73.9	100	100	92	91
u5_mort	10.3	33.0	30.3	-8.2	17.8	16.1	100	98	98	75
enrol_secondary	13.6	81.7	82.6	+1.1	88.3	91.4	87	94	91	78
internet	4.7	45.2	46.2	+2.0	44.7	46.2	100	98	96	91
co2_pc	7.0	4.87	4.94	+1.5	2.63	3.06	97	98	96	86
literacy	27.6	82.0	85.5	+4.3	92.0	94.6	97	90	83	44
poverty_190	44.9	19.2	13.4	-30.5	7.83	1.84	74	86	68	19
gini	32.2	38.6	37.8	-2.1	37.4	36.0	74	90	70	49
health_exp	12.1	6.79	6.83	+0.6	6.49	6.67	90	96	98	75

- For well-covered indicators (GDP, life expectancy, internet, CO_2; $<8\,\%$ missing) the mean moves by less than $3\,\%$: missing values barely matter.
- For **poverty**, the reporting-only mean ($19.2\,\%$) overstates world poverty because rich countries are absent; with imputation the mean falls by $30\,\%$ to $13.4\,\%$ and the median from $7.8\,\%$ to $1.8\,\%$.
- For **literacy** the bias goes the other way: rich countries are missing, so the reporting-only mean ($82.0\,\%$) is too low ($85.5\,\%$ after imputation). Under-5 mortality is $8\,\%$ too high for the same reason.
- Conclusion: summary statistics of sparse, survey-based indicators describe the countries that report, not the world. We therefore exclude indicators with $<60\,\%$ coverage from the main analysis and state the bias wherever they are used.

A2. Summary

- Income, population, CO_2, energy and mortality are strongly right-skewed (mean $\gg$ median), so we use medians and log scales; percentages are bounded and left-skewed.
- $44.9\,\%$ of panel cells are missing, mostly structurally. Missingness depends on income group (missing not at random).
- Impact: negligible ($<3\,\%$ shift of the mean) for well-covered indicators, but large for survey indicators: the poverty mean drops $30\,\%$ and the literacy mean rises $4\,\%$ once missing countries are accounted for.

2.3 A3. Exploration: outliers, clusters, interesting attributes

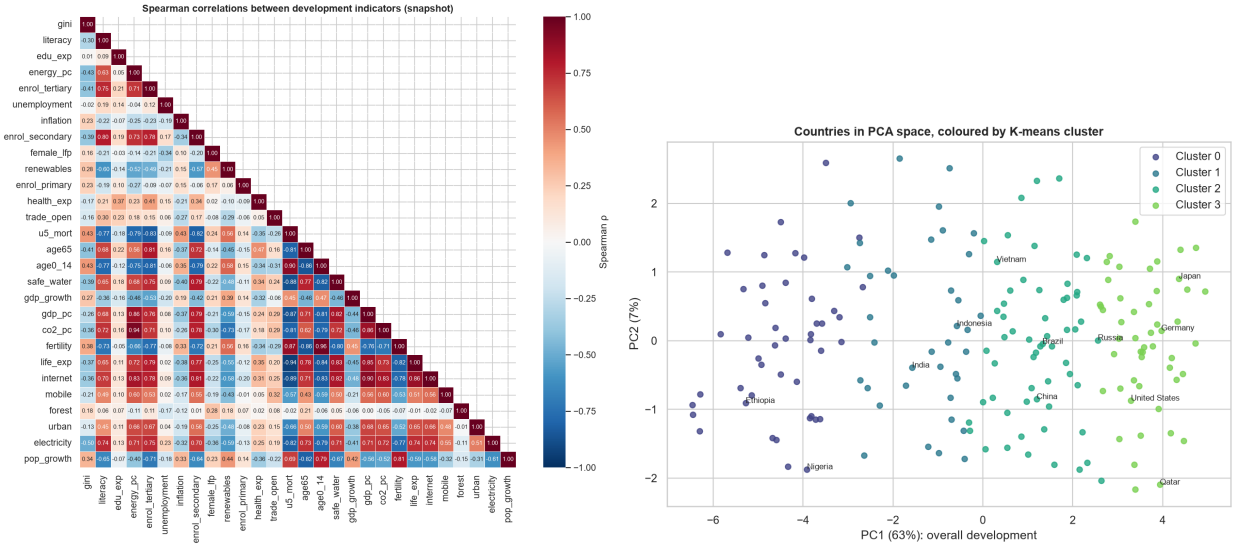
Outliers. We flagged values beyond 1.5 interquartile ranges from the quartiles, on a log scale for skewed indicators (Table 4). In the raw panel we also found 29 **hyperinflation** country-years above $500\,\%$ (DR Congo 1994: $23{,}773\,\%$; Bolivia 1985: $11{,}750\,\%$; Peru 1990: $7{,}482\,\%$).

Table 4: IQR outliers in the snapshot (indicators with at least one outlier).

Indicator	Scale	n	Outliers	Most extreme (value)
`gdp_growth`	raw	199	4	Libya (-24), Ukraine (-6.8), Aruba (-5.65)
`inflation`	raw	176	8	Venezuela (62.2), South Sudan (47.3), Sudan (36.9)
`unemployment`	raw	174	11	Mauritania (31), Macedonia (27.9), Bosnia and Herzegovina (27.9)
`trade_open`	raw	189	5	Hong Kong SAR, China (439), Luxembourg (374), Singapore (351)
`gini`	raw	145	4	South Africa (63.4), Namibia (61), Haiti (60.8)
`pop_growth`	raw	214	2	Oman (8.09), Andorra (-4.19)
`fertility`	raw	203	1	Niger (7.62)
`female_lfp`	raw	186	7	Syrian Arab Republic (13.6), Iraq (15), Algeria (15.4)
`health_exp`	raw	188	3	Tuvalu (19.7), United States (17.1), Marshall Islands (16.5)
`safe_water`	raw	199	17	Somalia (31.7), Papua New Guinea (40), Equatorial Guinea (47.8)
`enrol_primary`	raw	188	16	Somalia (29.2), Curaçao (165), Gabon (165)
`enrol_secondary`	raw	185	1	Belgium (166)
`literacy`	raw	155	6	Niger (15.5), Guinea (25.3), South Sudan (26.8)
`edu_exp`	raw	165	6	Lesotho (13), Cuba (12.8), Solomon Islands (9.85)
`co2_pc`	log10	199	1	Burundi (0.0213)
`energy_pc`	log10	168	3	Lesotho (9.72), Timor-Leste (60.2), Guinea-Bissau (62.2)
`mobile`	raw	205	3	Macao SAR, China (323), Hong Kong SAR, China (239), Kuwait (218)

Most outliers are *real* values, not errors: small trading hubs (Hong Kong, Luxembourg, Singapore) have trade above 300 % of GDP; Gulf states have very high CO_2; Libya's GDP fell 24 % in a war year; small island states spend heavily on health. Gross primary enrolment above 160 % (Gabon, Curaçao) reflects over-age pupils. Because the outliers are genuine, we keep them, use robust statistics (medians, Spearman correlation, log scales) and only *flag* hyperinflation years (B1).

Clusters of similar instances. Even before modelling, countries group clearly. The correlation matrix (Figure 4a) shows one large block: income, life expectancy, internet, CO_2, energy, water and schooling all correlate with GDP per capita at $|\rho| = 0.76$–0.90, and with each other. A PCA of 15 standardised indicators puts 63.5 % of the variance on the first component (Figure 4b), so countries lie along a single "development axis". Along that axis, Sub-Saharan Africa forms a distinct low group, and oil exporters form a group that is rich but has poor health outcomes.



(a) Spearman correlations **(b)** PCA projection with K-means clusters

Figure 4: Indicators move together, and countries spread along one main axis.

Interesting attributes and values.

- **Life expectancy** is the most informative outcome: it rose in every region, by up to 30 years in South Asia (39 → 69), and the gap between high- and low-income countries fell from 30.1 to 20.8 years.
- **Fertility** separates Sub-Saharan Africa from all other regions: it fell only 28 % there (6.7 → 4.8) against 60–67 % elsewhere; Niger (7.6) is the only fertility outlier.
- **GDP per capita** spans three orders of magnitude (US$153 to US$158,603); the p90/p10 ratio is about 80.
- **Vietnam** is an interesting value in almost every relationship: GDP per capita of US$1,078 but a life expectancy of 75.8 years, equal to China and Thailand, which are 3.5 times richer (Figure 5).
- Large negative residuals: Swaziland, Equatorial Guinea, Angola and South Africa live 15–20 years less than their income predicts (HIV and oil-based income).

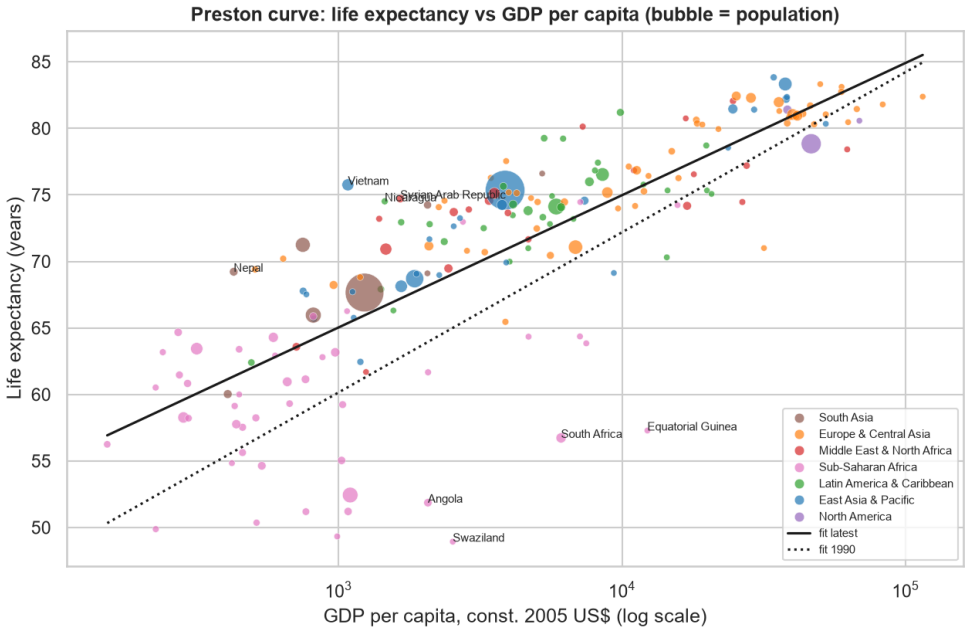


Figure 5: The Preston curve [3]: life expectancy vs log GDP per capita. Vietnam (red) is the furthest above the curve.

Impact of missing values on the exploration. Table 5 recomputes key correlations on the complete pairs and after imputation.

Table 5: Spearman ρ with pairwise deletion vs income-group median imputation.

Pair	n pairwise	ρ pairwise	n imputed	ρ imputed	$\Delta\rho$
`gdp_pc` ∼ `life_exp`	192	0.85	214	0.87	+0.01
`gdp_pc` ∼ `co2_pc`	192	0.86	214	0.85	-0.01
`gdp_pc` ∼ `literacy`	150	0.68	214	0.64	-0.04
`gdp_pc` ∼ `poverty_190`	118	-0.77	214	-0.79	-0.02
`gdp_pc` ∼ `gini`	145	-0.26	214	-0.42	-0.16
`life_exp` ∼ `health_exp`	182	0.35	214	0.38	+0.03

- For well-covered pairs the correlation changes by ≤ 0.03, so the exploration is robust.

- For sparse indicators imputation *changes the conclusion*: the Gini–income correlation goes from -0.26 to -0.42. Income-group medians are a function of income, so imputation artificially strengthens relationships with income. We therefore never use imputed values for correlation or regression on sparse indicators.
- Listwise deletion is not an option: only **69 of 214** countries report all 32 indicators, and 147 report all 15 clustering indicators. Dropping incomplete countries would remove mostly poor and small countries and bias every result towards rich countries. This motivates the strategy in B1.

A3. Summary

- Outliers are mostly genuine (trade hubs, Gulf CO_2, wars, hyperinflation); we keep them and use robust methods. 29 hyperinflation years are flagged.
- Clusters: indicators form one correlated block around income, and countries lie on one main axis (PC1 = 63.5 %), with Sub-Saharan Africa and oil exporters as distinct groups.
- Interesting attributes: life expectancy (big convergence), fertility (Sub-Saharan exception), GDP (huge spread), and Vietnam as the top over-performer.
- Missing values: little effect on well-covered correlations, but imputation biases sparse ones (Gini: $-0.26 \rightarrow -0.42$), and listwise deletion would keep only 69 countries.

3 Data preprocessing

Based on the exploration we applied all four classic groups of techniques [7], plus two project-specific ones. Table 6 summarises what each step changed.

Table 6: Effect of the preprocessing pipeline.

Step	Technique	Result
Reduction	chunked read, filter 32 of 1,344 indicators	5.66 M → 247,446 rows, 34.9 MB in memory
Cleaning	separate aggregates	210,443 country rows, 37,003 aggregate rows set aside
Cleaning	duplicates / non-finite / out-of-domain	0 / 0 / 0 found
Cleaning	flag hyperinflation (> 500 %)	29 country-years flagged, kept
Integration	join `Country.csv`, `Series.csv`	region, income group, names, topics attached
Transformation	long → wide	panel 11,935 country-years × 32
Cleaning	interpolation of gaps ≤ 3 years	missing cells 44.9 % → 42.1 %
Reduction	2005–2014 snapshot	214 countries × 32, median data year 2014
Reduction	coverage threshold 60 %	31 indicators kept (poverty excluded)
Transformation	$\log_{10}$, standardisation	symmetric, comparable features
Reduction	PCA (15 indicators)	PC1 63.5 %, PC1+PC2 70.8 %

3.1 B1. Data cleaning

- **Aggregates removed.** The 33 aggregates (World, regions, income groups) would double-count countries; they are identified by an empty `Region` and set aside.
- **Validation.** No duplicate (country, indicator, year) keys, no infinite values, and no values outside the valid domain (e.g. percentages within 0–100 where applicable).
- **Outliers.** Genuine extremes are kept (A3). The 29 hyperinflation years are flagged rather than deleted, and inflation is log-transformed for modelling.
- **Missing values**, handled according to their type:

1. interpolate only short internal gaps (≤ 3 years, linear, never extrapolate): 44.9 % $\rightarrow$ 42.1 % missing;
2. for the cross-country snapshot, take the latest value in a 10-year window (2005–2014);
3. impute income-group medians only inside models and clustering (fitted inside each cross-validation fold, so no information leaks from test countries);
4. exclude indicators below 60 % coverage from the analysis (poverty at \$1.90, 55 %);
5. plot group trends only for years where at least 30 % of the group's countries report.

Result: no country is dropped because of a few gaps (192 countries enter the models instead of 69 under listwise deletion), and E1.3 (Section 4) checks that the conclusions do not depend on the changing set of reporting countries.

3.2 B2. Data integration

The three files are joined on their keys: `Indicators.CountryCode` $\rightarrow$ `Country.CountryCode` adds `ShortName`, `Region` and `IncomeGroup`; `Indicators.IndicatorCode` $\rightarrow$ `Series.SeriesCode` adds the indicator name, topic and unit. The two high-income groups (OECD and non-OECD) are merged into one "High income" level, giving an ordinal income attribute with four levels. Every indicator receives a short alias (e.g. `SP.DYN.LE00.IN` $\rightarrow$ `life_exp`) and a theme.

3.3 B3. Data transformation

- **Reshaping:** long table $\rightarrow$ wide country-year panel (one column per indicator) and a wide country snapshot.
- **Log transform** ($\log_{10}$) of right-skewed attributes: GDP per capita, CO_2, energy, trade, inflation (signed log), mortality and population. This makes relationships close to linear (e.g. the Preston curve) and reduces the leverage of outliers.
- **Standardisation** (z-scores) before PCA, clustering, distance computations and linear models, fitted inside each cross-validation fold.
- **Derived attributes:** annual growth rates (CAGR, log growth) for convergence analysis; country and year means for the fixed-effects model; out-of-fold residuals as "performance vs income".

3.4 B4. Data reduction

- **Attribute selection:** 1,344 $\rightarrow$ 32 indicators (domain knowledge + coverage), then 31 after the coverage threshold; 15 for clustering; 10 social indicators without income for the "development twins" (E3.2).
- **Instance reduction:** chunked filtering reduced 5.66 M rows to 247k without loading the 574 MB file at once; the snapshot reduces 55 years to one value per country.
- **Dimensionality reduction:** PCA on 15 standardised indicators: PC1 explains 63.5 % and PC2 7.3 %. PC1 loads almost equally on income, health, schooling, internet and CO_2, which confirms a single development dimension; PC2 is mostly forest cover.

3.5 B5. Other techniques

- **Leakage control.** For each model target we drop near-definitional features: infant and under-5 mortality when predicting life expectancy, age structure and population growth when predicting fertility, PPP GDP and GDP growth when predicting GDP, primary and tertiary

enrolment when predicting secondary enrolment, energy use when predicting CO_2, mobile when predicting internet.

- **Grouped cross-validation.** Folds are split by country, so the same country never appears in training and test data (in the panel, its other years would otherwise leak).
- **Balanced panel and population weights** (E1.3) as a robustness check against changing coverage.

4 Data mining actions

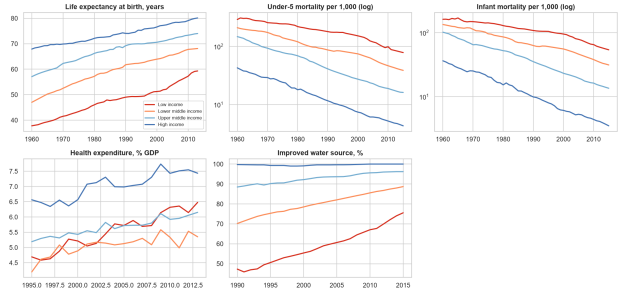
Table 7 lists the data-mining tasks. Items 1–6 come from the analysis notebook; E1.1–E3.3 are the nine storyline experiments implemented in `wdi/story.py`.

Table 7: Data-mining actions.

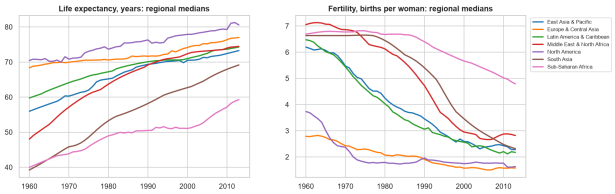
#	Task	Method	Question
1	Trend analysis	group medians by income group and region	How did each theme change since 1960?
2	Inequality & convergence	Gini, p90/p10, σ- and β-convergence (E1.1–E1.3)	Are countries becoming more alike?
3	Association	Spearman correlation, Preston curve, Kuznets curve	What moves with income?
4	Prediction (regression)	19 regressors × 6 targets, grouped 5-fold CV, 3–5 seeds	How predictable is each theme?
5	Residual & panel analysis	income-only models (E2.1, E2.2), two-way fixed effects (E2.3)	How much does income explain, and who beats it?
6	Clustering	K-means, Gaussian mixture, Ward; PCA (E3.1)	Do data-driven groups match income groups?
7	Similarity search	nearest neighbours in social-indicator space (E3.2)	Which countries is Vietnam really like?
8	Gap analysis	standardised distance to cluster medians (E3.3)	What separates Vietnam from the next group?

4.1 Trends by theme

Group-median trends (Figure 6) show broad social progress. Life expectancy rose everywhere; primary enrolment became near-universal; mobile subscriptions reached about 64 per 100 even in low-income countries. Median GDP per capita grew 2.1 %/yr in upper-middle and 1.7 %/yr in lower-middle income countries over 1990–2014, against 0.6 %/yr in high-income countries.



(a) Health



(b) Regions: life expectancy and fertility

Figure 6: Group-median trends. All regions gained life expectancy; fertility stalled only in Sub-Saharan Africa.

4.2 Convergence (E1.1–E1.3)

E1.1 σ-convergence measures the cross-country coefficient of variation (CV) each year. Social indicators converge strongly; income does not move (Figure 7, Table 8).

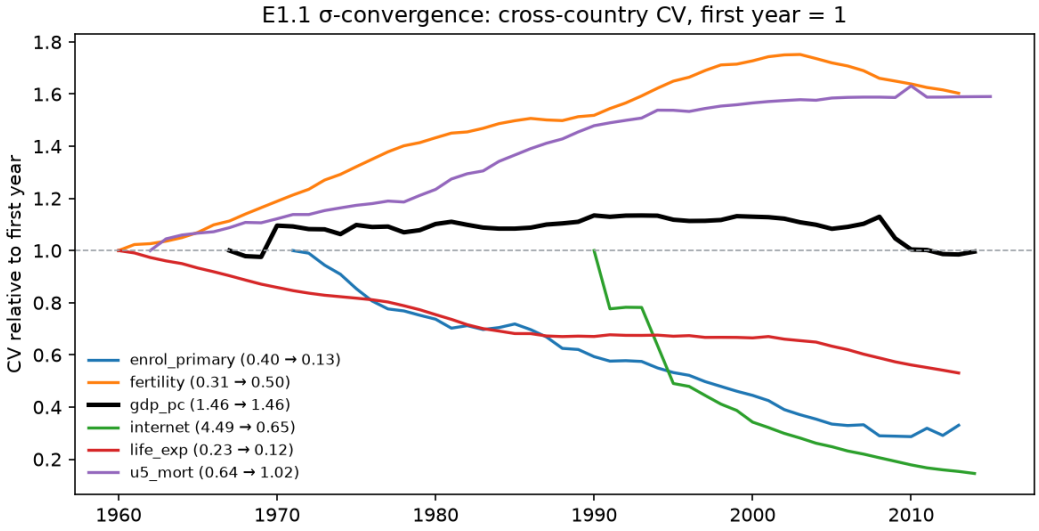


Figure 7: E1.1: cross-country CV relative to the first year. Health, schooling and internet converge; GDP stays flat; fertility diverges.

Table 8: E1.1 σ-convergence (years with >100 reporting countries).

Indicator	First year	CV	Last year	CV	Change %
`life_exp`	1960	0.23	2013	0.12	-47
`enrol_primary`	1971	0.40	2013	0.13	-67
`gdp_pc`	1967	1.46	2014	1.46	-0
`fertility`	1960	0.31	2013	0.50	+60
`u5_mort`	1962	0.64	2015	1.02	+59
`internet`	1990	4.49	2014	0.65	-85

Under-5 mortality "diverges" only in relative terms: as rich countries approach zero, the ratio between countries grows even though absolute gaps shrink.

E1.2 β-convergence [4] regresses the annual change since 1990 on the 1990 level (Figure 8, Table 9). A negative β means that countries starting lower improve faster.

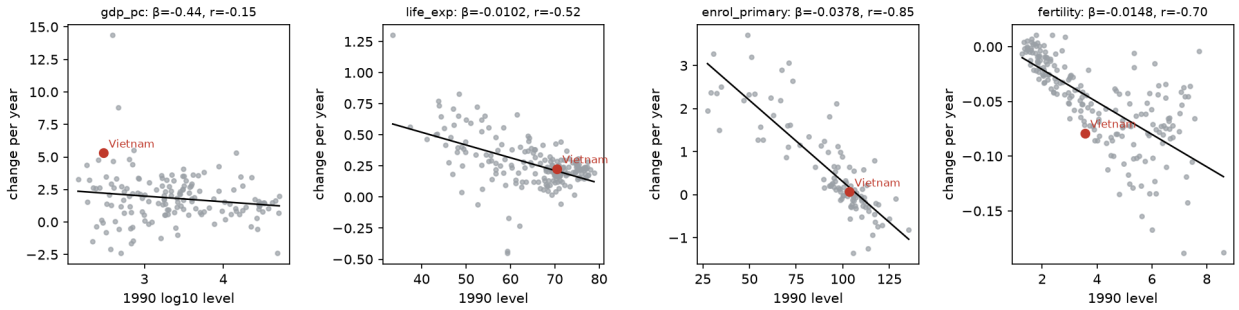


Figure 8: E1.2: change per year vs 1990 level. Vietnam (red) grew far faster than its 1990 income predicts.

Table 9: E1.2 β-convergence.

Indicator	Period	n	β	95% CI	r	p	Vietnam change/yr
gdp_pc	1990-2014	157	-0.440	[-0.888, 0.008]	-0.15	0.054	5.31
life_exp	1990-2013	195	-0.010	[-0.013, -0.008]	-0.52	<0.001	0.23
enrol_primary	1990-2013	105	-0.038	[-0.042, -0.033]	-0.85	<0.001	0.07
fertility	1990-2013	195	-0.015	[-0.017, -0.013]	-0.70	<0.001	-0.08

E1.3 robustness. Because poorer countries report less, convergence could be an artefact of countries entering the data. We recompute the CV change on a balanced panel and with population weights (Table 10).

Table 10: E1.3: CV change 1990 → 2013 on three samples.

Indicator	Sample	n (1990/2013)	CV 1990	CV 2013	Change %	Direction
life_exp	all reporting	200/198	0.15	0.12	-21	converging
life_exp	balanced panel	195/195	0.15	0.12	-22	converging
life_exp	balanced, population-weighted	195/195	0.12	0.10	-20	converging
enrol_primary	all reporting	160/119	0.24	0.13	-44	converging
enrol_primary	balanced panel	105/105	0.26	0.12	-52	converging
enrol_primary	balanced, population-weighted	105/105	0.20	0.12	-40	converging
gdp_pc	all reporting	172/188	1.66	1.44	-13	converging
gdp_pc	balanced panel	166/166	1.53	1.45	-5	converging
gdp_pc	balanced, population-weighted	166/166	1.78	1.64	-8	converging
fertility	all reporting	199/198	0.47	0.50	+6	diverging
fertility	balanced panel	195/195	0.48	0.50	+5	flat
fertility	balanced, population-weighted	195/195	0.45	0.48	+7	diverging
u5_mort	all reporting	192/192	0.95	1.02	+8	diverging
u5_mort	balanced panel	192/192	0.95	1.02	+8	diverging
u5_mort	balanced, population-weighted	192/192	0.75	0.90	+21	diverging
internet	all reporting	204/203	4.49	0.69	-85	converging
internet	balanced panel	200/200	4.44	0.69	-84	converging
internet	balanced, population-weighted	200/200	3.55	0.68	-81	converging

4.3 How much does income explain? (E2.1–E2.3)

E2.1 splits the cross-validated R^2 into three layers per theme: ① Ridge on log GDP only, ② Ridge on all non-leaking features, ③ the best of the 19-model suite (Figure 9, Table 11).

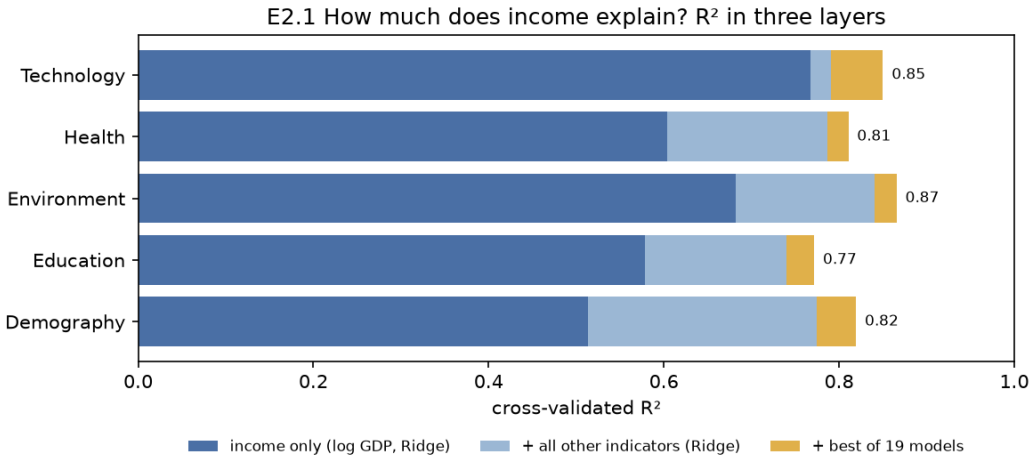


Figure 9: E2.1: income alone (dark) carries most of the explained variance; complex models (yellow) add little.

Table 11: E2.1 cross-validated R^2 (mean ± std over 5 seeds, snapshot, 5-fold country-grouped CV).

Theme	① Income only	② All features	③ Best model	Best
Demography	0.51 ± 0.007	0.77 ± 0.015	0.82 ± 0.009	Voting
Education	0.58 ± 0.005	0.74 ± 0.006	0.77 ± 0.007	Stacking
Environment	0.68 ± 0.004	0.84 ± 0.007	0.87 ± 0.009	Voting
Health	0.60 ± 0.004	0.79 ± 0.009	0.81 ± 0.008	Voting
Technology	0.77 ± 0.002	0.79 ± 0.009	0.85 ± 0.003	ExtraTrees
Mean	0.63	0.79	0.82	

E2.2 residual scorecard. The out-of-fold residual of the income-only model tells how much better or worse a country does than its income predicts (Figure 10, Table 12).

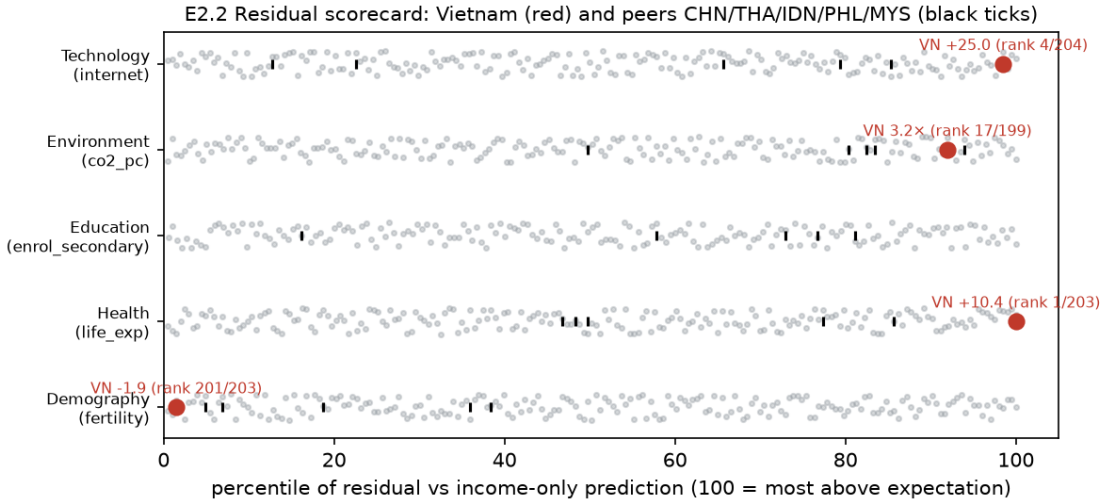


Figure 10: E2.2: percentile of each country's residual. Vietnam is first of 203 for life expectancy, fourth for internet, and emits 3.2× its predicted CO_2.

Table 12: E2.2 residuals for Vietnam and peers (Vietnam does not report secondary enrolment for 2005–2014). For log targets the residual is actual/predicted.

Theme	Target	Country	Actual	Predicted	Residual	Rank
Demography	`fertility`	Philippines	3.01	3.41	-0.4	126/203
Demography	`fertility`	Malaysia	1.96	2.41	-0.5	131/203
Demography	`fertility`	Indonesia	2.48	3.31	-0.8	166/203
Demography	`fertility`	China	1.67	2.84	-1.2	190/203
Demography	`fertility`	Thailand	1.52	2.86	-1.3	194/203
Demography	`fertility`	**Vietnam**	1.74	3.69	-1.9	201/203
Education	`enrol_secondary`	Philippines	85.4	69.3	+16.1	36/185
Education	`enrol_secondary`	Indonesia	83.1	70.2	+12.9	44/185
Education	`enrol_secondary`	China	92.4	80.9	+11.5	51/185
Education	`enrol_secondary`	Thailand	85.9	80.6	+5.4	79/185
Education	`enrol_secondary`	Malaysia	70.8	90.1	-19.3	156/185
Environment	`co2_pc`	China	6.71	1.88	3.6×	13/199
Environment	`co2_pc`	**Vietnam**	1.97	0.612	3.2×	17/199
Environment	`co2_pc`	Thailand	4.53	1.86	2.4×	34/199
Environment	`co2_pc`	Malaysia	7.90	3.33	2.4×	36/199
Environment	`co2_pc`	Indonesia	2.30	0.996	2.3×	40/199
Environment	`co2_pc`	Philippines	0.868	0.910	1.0×	101/199
Health	`life_exp`	**Vietnam**	75.8	65.4	+10.4	1/203
Health	`life_exp`	China	75.4	71.0	+4.4	30/203
Health	`life_exp`	Thailand	74.2	70.8	+3.4	47/203
Health	`life_exp`	Malaysia	74.6	73.7	+0.9	103/203
Health	`life_exp`	Indonesia	68.7	67.9	+0.8	106/203
Health	`life_exp`	Philippines	68.1	67.5	+0.7	109/203
Technology	`internet`	**Vietnam**	48.3	23.3	+25.0	4/204
Technology	`internet`	Malaysia	67.5	54.5	+13.0	31/204
Technology	`internet`	Philippines	39.7	29.6	+10.1	43/204
Technology	`internet`	China	49.3	43.5	+5.8	71/204
Technology	`internet`	Thailand	34.9	43.1	-8.2	159/204
Technology	`internet`	Indonesia	17.1	31.8	-14.6	179/204

Across all countries the largest positive life-expectancy residual is Vietnam's; the largest negative ones are Swaziland, Equatorial Guinea and Angola.

E2.3 fixed effects. The Preston curve compares countries. To see what happens when a country *gets* richer, we estimate a two-way fixed-effects model [8] (country and year effects, 1990–2014) with country-clustered standard errors (Table 13, Figure 11).

Table 13: E2.3: life expectancy on $\log_{10}$ GDP per capita.

Model	Years per 10× GDP pc	SE	95% CI	Obs.	Countries
Between countries (country means)	10.89	0.56	[9.79, 11.98]	196	196
Pooled OLS	11.09	0.51	[10.10, 12.09]	4,377	196
Two-way fixed effects (country + year)	3.24	1.00	[1.28, 5.20]	4,377	196

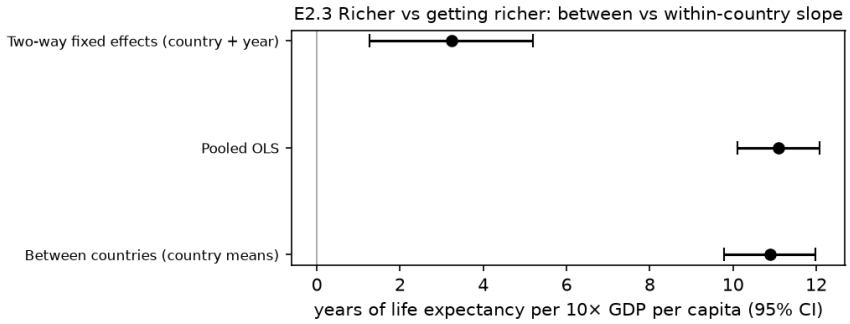


Figure 11: E2.3: within-country slope is about one third of the between-country slope.

4.4 Clustering, similarity and gaps (E3.1–E3.3)

Clustering. K-means on 15 standardised indicators (192 countries with $\geq 70\,\%$ coverage, 2.5 % of cells imputed). Silhouette is highest at $k = 2$ (0.40) and falls to 0.19 at $k = 4$ (Figure 12); we use $k = 4$ for interpretability and comparability with the four income groups. Clusters are named by median GDP: low development, transition, emerging, advanced.

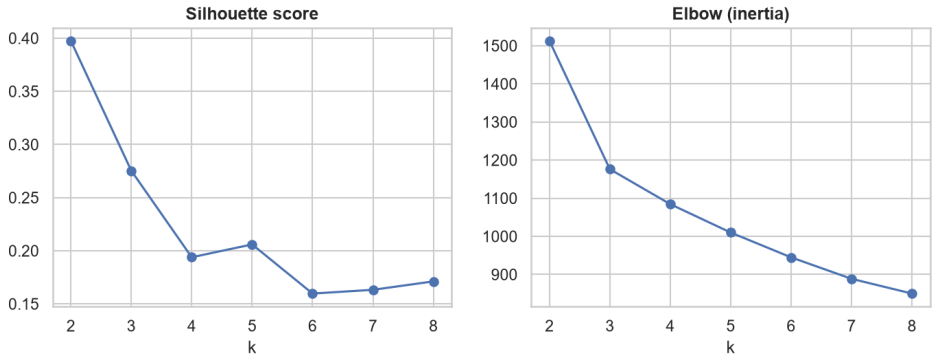


Figure 12: Choice of k: elbow and silhouette. Low silhouettes suggest a continuum rather than well-separated groups.

E3.1 robustness. We repeat clustering with three algorithms and 20 seeds each (Table 14, Figure 13).

Table 14: E3.1: stability of the partition (reference = K-means, seed 42, as in the app).

Algorithm	ARI between seeds	ARI vs reference	Agreement w/ income	Vietnam 'Emerging'
Agglomerative (Ward)	1.00	0.55	65%	0%
Gaussian mixture	0.70	0.54	63%	35%
K-means	0.93	0.95	74%	90%

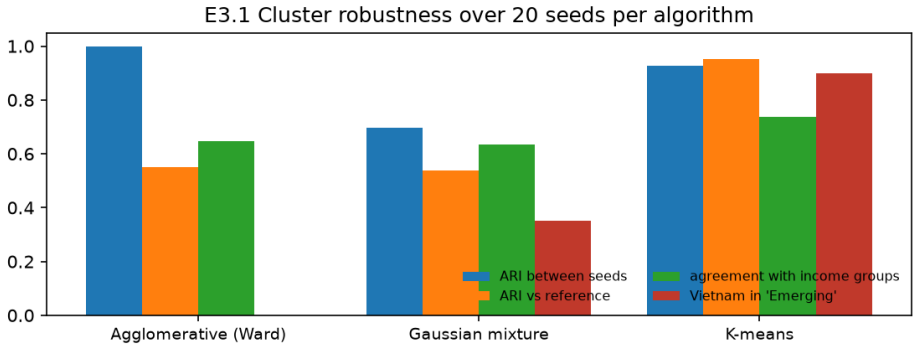


Figure 13: E3.1: K-means is stable across seeds, but the partition depends on the algorithm.

E3.2 development twins. Nearest neighbours of Vietnam by GDP per capita alone vs by 10 standardised social indicators without income (Table 15).

Table 15: E3.2: Vietnam's five nearest countries in two spaces.

Rank	Same income	Life exp.	Same social outcomes	Income vs VN	Life exp.
1	Côte d'Ivoire	51.2	Seychelles	14.5×	74.2
2	São Tomé and Principe	66.3	Mauritius	6.6×	74.5
3	Nigeria	52.4	Thailand	3.5×	74.2
4	Solomon Islands	67.7	Grenada	6.2×	73.2
5	Zambia	59.2	St. Lucia	5.2×	74.9

E3.3 gap analysis. Difference between Vietnam and the median of the "emerging" and "advanced" clusters, in standard deviations across countries (Table 16, Figure 14).

Table 16: E3.3: Vietnam vs cluster medians (gap in SD; log scale where noted). Secondary enrolment is omitted because Vietnam does not report it.

Indicator	Vietnam	Emerging median	Gap (SD)	Advanced median	Gap (SD)
`trade_open` (log)	170	82.2	+1.52	102	+1.06
`urban`	33.0	66.1	-1.42	81.5	-2.08
`gdp_pc` (log)	1,078	5,306	-1.03	27,960	-2.10
`forest`	46.8	31.0	+0.67	32.1	+0.64
`energy_pc` (log)	731	1,430	-0.57	3,693	-1.28
`mobile`	147	127	+0.47	123	+0.55
`co2_pc` (log)	1.97	3.88	-0.42	7.50	-0.83
`u5_mort` (log)	22.3	14.6	+0.38	4.20	+1.45
`fertility`	1.74	2.00	-0.18	1.60	+0.10
`life_exp`	75.8	74.3	+0.17	80.6	-0.57
`age65`	6.62	7.42	-0.13	16.0	-1.62
`internet`	48.3	50.3	-0.04	81.5	-1.14
`electricity`	99.0	100	-0.03	100	-0.03
`safe_water`	96.4	96.2	+0.01	100	-0.24

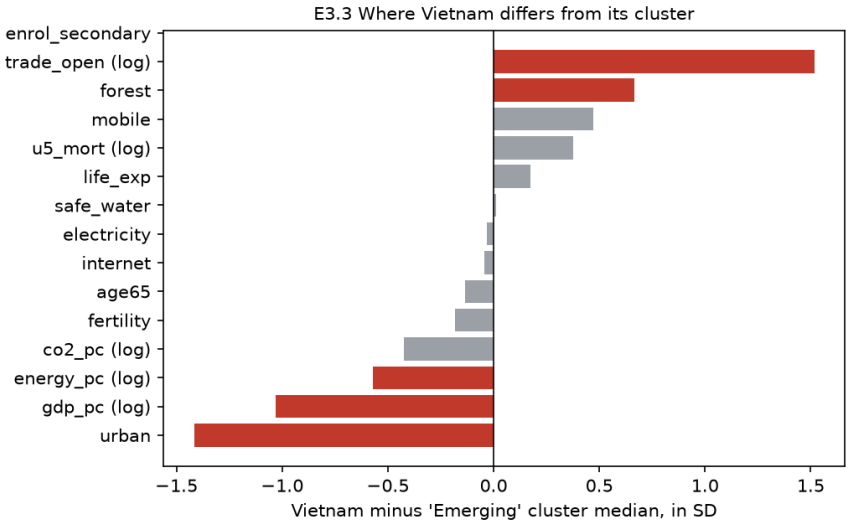


Figure 14: E3.3: Vietnam is behind the emerging cluster on urbanisation and income, level on health and technology, ahead on trade openness and forest cover.

5 Metrics and evaluation

5.1 Evaluation protocol

- **Regression metrics:** coefficient of determination $R^2 = 1 - \sum(y - \hat{y})^2 / \sum(y - \bar{y})^2$, mean absolute error (MAE) and root mean squared error (RMSE), always computed on *out-of-fold* predictions.
- **Validation:** 5-fold `GroupKFold` by country; repeated with 5 seeds (snapshot) or 3 seeds (panel); we report mean ± standard deviation across seeds. Imputation and scaling are fitted inside each training fold.
- **Clustering metrics:** silhouette coefficient [9] (separation), adjusted Rand index [10] (ARI, agreement between two partitions, 1 = identical, 0 = chance), and agreement with World Bank income groups (share of countries in their cluster's dominant income group).
- **Statistical inference:** slopes with 95 % confidence intervals and p-values; country-clustered standard errors in the panel.

5.2 Model suite results

Each theme has one target predicted from the other indicators (leaking features removed). Table 17 compares the linear baseline, random forest and the best of the 19 models; Figure 17a in Section 6 shows all models.

Table 17: Model suite: cross-validated R^2, and MAE/RMSE of the best model in target units ($\log_{10}$ for GDP and CO_2; years for life expectancy; births per woman for fertility; percentage points for enrolment and internet).

Data	Theme	Linear R^2	RF R^2	Best model	Best R^2	MAE	RMSE
Snapshot	Demography	0.778	0.796	Voting	0.819 ± 0.009	0.434	0.595
Snapshot	Economy	0.791	0.861	ExtraTrees	0.876 ± 0.008	0.179	0.243
Snapshot	Education	0.729	0.756	Stacking	0.772 ± 0.007	10.755	13.705
Snapshot	Environment	0.834	0.851	Voting	0.866 ± 0.009	0.186	0.252
Snapshot	Health	0.783	0.784	Voting	0.811 ± 0.008	2.692	3.689
Snapshot	Technology	0.787	0.841	ExtraTrees	0.850 ± 0.003	8.566	11.367
Panel	Demography	0.796	0.832	Voting	0.854 ± 0.004	0.494	0.660
Panel	Economy	0.809	0.836	Stacking	0.874 ± 0.007	0.183	0.252
Panel	Education	0.776	0.782	Stacking	0.811 ± 0.008	10.502	13.733
Panel	Environment	0.836	0.839	Voting	0.866 ± 0.001	0.192	0.272
Panel	Health	0.817	0.826	Voting	0.851 ± 0.002	2.708	3.815
Panel	Technology	0.691	0.888	ExtraTrees	0.901 ± 0.004	4.506	7.730

- All six themes are well predicted (R^2 0.77–0.90). Life expectancy is predicted within 2.7 years on average (MAE).
- **Ensembles win** (Voting, Stacking, ExtraTrees), but the margin over a plain linear model is small: 0.03–0.09 R^2 in every snapshot theme. The exception is internet in the panel (0.69 → 0.90), where adoption follows an S-curve over time that a linear model cannot fit.
- Seed variance of the best models is small (std ≤ 0.009), so differences above about 0.02 are meaningful.
- In the notebook's simple life-expectancy model, log GDP alone gives R^2 0.64 (MAE 3.6 years), seven extra indicators raise it to 0.76 (MAE 2.9), and a random forest [11] gives 0.76 (MAE 2.8): the relationship is essentially linear in log income, so the interpretable model is preferred.

Table 18: Permutation importance (drop in R^2 on 25 % held-out countries, best snapshot model).

Theme	Model	Top-3 features (permutation importance, drop in R^2)
Economy	ExtraTrees	`internet` (0.17), `co2_pc` (0.13), `life_exp` (0.03)
Demography	Voting	`u5_mort` (0.12), `life_exp` (0.06), `electricity` (0.03)
Health	Voting	`fertility` (0.16), `electricity` (0.09), `gdp_pc` (0.06)
Education	Stacking	`age0_14` (0.09), `electricity` (0.03), `u5_mort` (0.03)
Environment	Voting	`gdp_pc_ppp` (0.25), `renewables` (0.15), `internet` (0.07)
Technology	ExtraTrees	`gdp_pc` (0.16), `u5_mort` (0.04), `infant_mort` (0.04)

Feature importance (Table 18) confirms that development indicators substitute for each other: income is best predicted by internet and CO_2, life expectancy by fertility and electricity, CO_2 by income.

5.3 Evaluation of the experiments

- **Convergence (E1.1–E1.3).** Accepted: social indicators converge (life-expectancy CV −47 %, primary enrolment −67 %, internet −85 %) while GDP is flat (1.46 → 1.46). β is strongly negative and significant for health, schooling and fertility ($r = -0.52$ to -0.85, $p < 0.001$) but weak for GDP ($r = -0.15$, $p = 0.054$). The directions hold on a balanced panel and with population weights. One correction: more than half of GDP's small CV decline since 1990 is a *composition effect* (−13 % with all reporters, −5 % on the balanced panel).
- **Income share (E2.1).** Income alone gives R^2 0.51–0.77 (mean 0.63), about three quarters of what the best model reaches (mean 0.82). Other indicators add 0.16 and model complexity only 0.03.

- **Residuals (E2.2).** Vietnam is the largest positive life-expectancy residual of 203 countries (+10.4 years), fourth for internet (+25 per 100), and third-lowest fertility relative to income (−1.9). But it emits 3.2× the CO_2 its income predicts (rank 17 of 199).
- **Within vs between (E2.3).** A 10× income difference between countries goes with +10.9 years; within a country over time only +3.2 years (95 % CI 1.3–5.2). Most of the gain over time is common to all countries (year effects: medicine, vaccines), which is why the Preston curve shifted up about 5 years.
- **Clusters (E3.1).** K-means is stable (ARI 0.93 between seeds) and agrees with income groups for 74 % of countries. Across algorithms the partition changes (ARI about 0.55), and 77 of 192 countries stay in the same cluster in fewer than 60 % of the 60 runs. This, with the low silhouette, supports the reading of development as a continuum. Vietnam is "emerging" in 90 % of K-means runs but 35 % of Gaussian-mixture runs and never with Ward, so it sits on the boundary between transition and emerging.
- **Twins and gaps (E3.2–E3.3).** By income Vietnam resembles Côte d'Ivoire, Nigeria and Zambia (life expectancy 51–68). By social outcomes it resembles Seychelles, Mauritius and Thailand, which are 3.5–14.5 times richer. Against the emerging cluster, Vietnam's largest gaps are urbanisation (33 vs 66 %, −1.4 SD) and income (−1.0 SD); under-5 mortality (22 vs 15 per 1,000) is a moderate gap that becomes large (+1.4 SD) against advanced economies.

6 Interactive visualization web app

All results are also available in an interactive Streamlit app (`app.py`). It reads gzipped copies of the processed data (`app_data/`, 1.5 MB) and the precomputed model results (`results/`), so it runs without the raw Kaggle files:

```
pip install -r requirements.txt && streamlit run app.py
```

The app has five pages in a top navigation bar and a light/dark switch (top right). Every chart is an interactive Plotly figure (hover, zoom, legend filtering).

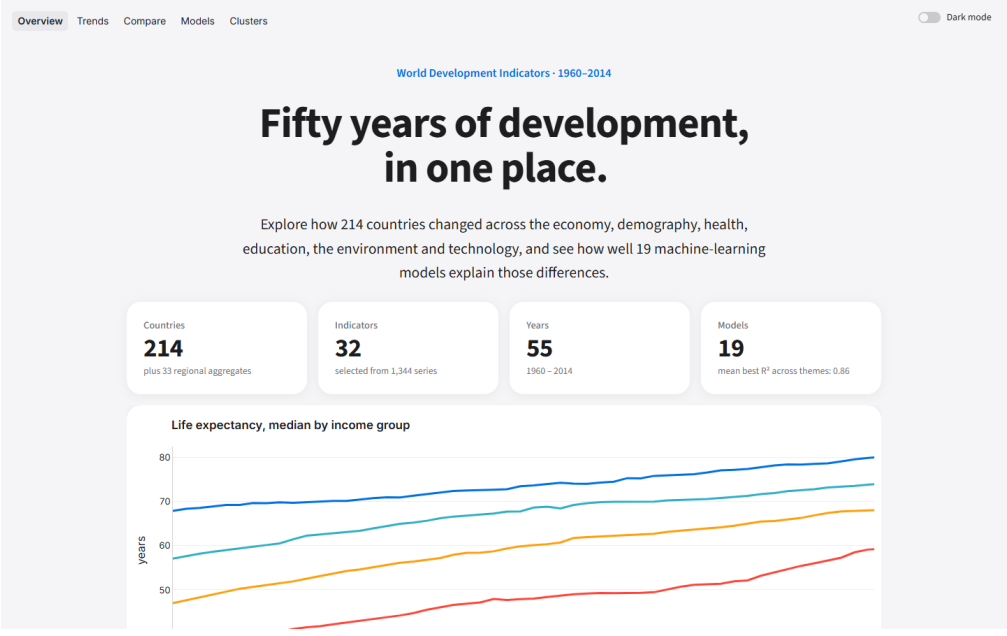
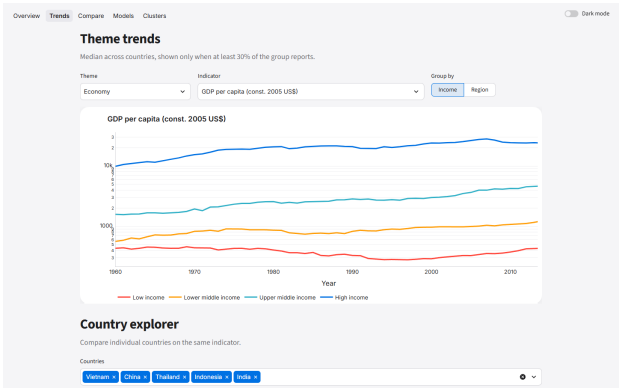
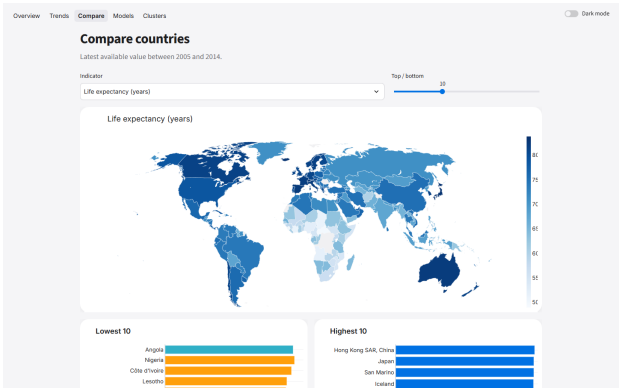


Figure 15: Overview: key numbers as tiles, life expectancy by income group, and the main insights.

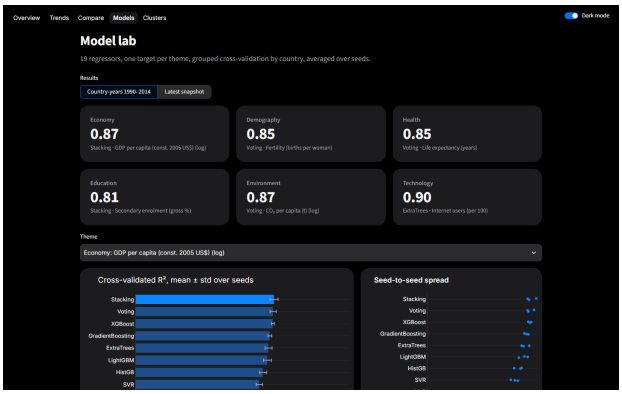


(a) Trends: any indicator by income group or region, plus a country comparison.

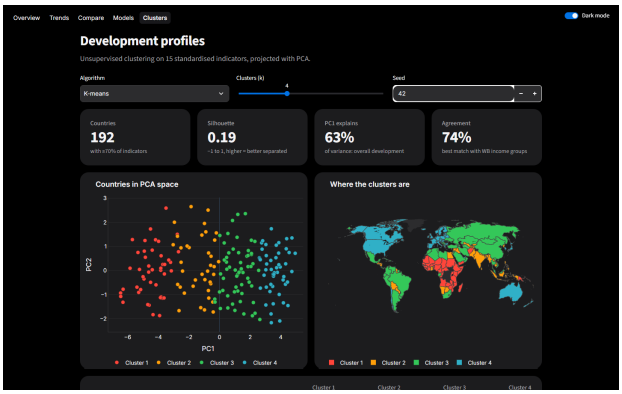


(b) Compare: world map, rankings, and a scatter of any two indicators.

Figure 16: Exploration pages.



(a) Models: R^2 of 19 models per theme with seed error bars, feature importance, and live experiments.



(b) Clusters: choose K-means, Gaussian mixture or Ward, k and seed; PCA, map and cluster profiles.

Figure 17: Modelling pages (dark mode).

- **Overview** summarises the dataset (countries, indicators, years) and the headline findings.
- **Trends** plots any of the 32 indicators as group medians (income group or region, with the 30 % coverage rule) and compares chosen countries.
- **Compare** shows a choropleth world map, top/bottom rankings, and a "relationship explorer" to scatter any two indicators on linear or log axes (used for the Preston and Kuznets curves).
- **Models** shows the precomputed suite results (snapshot and panel), permutation importance, actual vs predicted trends, and a *live experiment* panel where the user picks themes, models, seeds, folds, imputer and coverage threshold and re-runs the cross-validation in the browser.
- **Clusters** lets the user change the algorithm, k and seed and immediately see the silhouette, PCA projection, world map, cluster profiles and agreement with income groups. It uses the same `cluster_matrix`/`fit_clusters` functions as experiment E3.1, so the app and the report always agree (74 % at the default settings).

The app is also the presentation medium: the planned story pages (Part 1 What, Part 2 Why, Part 3 So what) reuse these pages and the experiment results in `results/story/` (see `docs/tasks.md`).

7 Key findings

1. **Broad social progress, led by health.** Life expectancy rose in every region (South Asia +30 years); the rich–poor gap fell from 30.1 to 20.8 years; cross-country dispersion of life expectancy halved and that of primary enrolment fell by two thirds.
2. **Income did not converge.** Middle-income countries grew faster than rich ones, but income dispersion did not change (CV 1.46 → 1.46; Gini 0.70 → 0.67; p90/p10 77× → 81×). Catch-up came from specific countries: China (9.2 %/yr), Vietnam (5.3–5.5 %), Lao PDR, India.
3. **Sub-Saharan Africa is the outlier in the demographic transition.** Fertility fell 28 % there against 60–67 % elsewhere, so fertility diverged across countries.
4. **Income explains most, but not all.** Income alone explains about 60 % of the variance in the other themes; ten times richer goes with about ten more years of life. But within a country, getting richer goes with only about three more years: most of the gain over time is shared by all countries.
5. **The residual is the story.** Vietnam, Nepal, Nicaragua and Bangladesh live 7–10 years longer than their income predicts; HIV-affected Southern Africa and oil economies live 15–20 years less.
6. **Growth still means more CO_2.** CO_2 correlates with income as tightly as life expectancy (ρ 0.86 vs 0.85); the environmental Kuznets [5] turning point is about US$51,000 and only 9 of 192 countries are above it; only 26 of 56 high-income countries cut per-capita CO_2 between 1990 and 2011.
7. **Development is one continuous axis.** PC1 explains 63.5 %; four clusters match income groups for 74 % of countries, with informative mismatches (oil-rich Equatorial Guinea and Angola in the low cluster), but clusters are not robust across algorithms.
8. **Vietnam** is lower-middle income with the social outcomes of countries 3.5–14.5 times richer, but it emits 3.2 times the CO_2 its income predicts. Its next gaps are income, urbanisation and child mortality.

8 Limitations and future work

- **Data age.** The Kaggle extract ends in 2014–2015 (CO_2 in 2011) and uses old income groups and 2005 prices.
- **Missing not at random.** Poorer countries report less (and rich countries do not report poverty). We handled this with interpolation, snapshots, in-fold imputation and robustness checks, but survey indicators remain biased.
- **Association, not causation.** Even the fixed-effects model only removes time-invariant country differences and common shocks; it does not identify causal effects of income.
- **Unweighted medians.** Tuvalu counts as much as China (a "typical country" view). E1.3 shows that population weighting does not change the convergence conclusions.
- **Territorial CO_2** ignores emissions embodied in trade.
- **Clustering** has low silhouettes and depends on the algorithm, so cluster labels should be read as positions on a continuum.

Future work: refresh from the live WDI API to cover the 2020s; dynamic panel models; population-weighted and consumption-based CO_2 analyses; build the three story pages of the app (`docs/tasks.md`).

9 Conclusion

Richer mostly means more developed: income alone explains about 60 % of cross-country differences in health, education, demography, environment and technology, and complex models add little to that. But income is not destiny. Health and schooling converged across the world while income did not, most of the gain over time was shared by all countries, and some countries, Vietnam above all, reach outcomes far above their income. The practical message is to judge development by outcomes, not by GDP alone, and, for Vietnam, to keep its social lead while closing the income, urbanisation and child-mortality gaps on a lower-carbon path than the countries that went before it.

A Appendix

A.1 Selected indicators

Table 19: The 32 selected indicators.

Alias	WDI code	Theme	Alias	WDI code	Theme
gdp_pc	NY.GDP.PCAP.KD	Economy	infant_mort	SP.DYN.IMRT.IN	Health
gdp_pc_ppp	NY.GDP.PCAP.PP.KD	Economy	u5_mort	SH.DYN.MORT	Health
gdp_growth	NY.GDP.MKTP.KD.ZG	Economy	health_exp	SH.XPD.TOTL.ZS	Health
inflation	FP.CPI.TOTL.ZG	Economy	safe_water	SH.H2O.SAFE.ZS	Health
unemployment	SL.UEM.TOTL.ZS	Economy	enrol_primary	SE.PRM.ENRR	Education
trade_open	NE.TRD.GNFS.ZS	Economy	enrol_secondary	SE.SEC.ENRR	Education
gini	SI.POV.GINI	Economy	enrol_tertiary	SE.TER.ENRR	Education
poverty_190	SI.POV.DDAY	Economy	literacy	SE.ADT.LITR.ZS	Education
population	SP.POP.TOTL	Demography	edu_exp	SE.XPD.TOTL.GD.ZS	Education
pop_growth	SP.POP.GROW	Demography	co2_pc	EN.ATM.CO2E.PC	Environment
urban	SP.URB.TOTL.IN.ZS	Demography	energy_pc	EG.USE.PCAP.KG.OE	Environment
fertility	SP.DYN.TFRT.IN	Demography	forest	AG.LND.FRST.ZS	Environment
age65	SP.POP.65UP.TO.ZS	Demography	renewables	EG.FEC.RNEW.ZS	Environment
age0_14	SP.POP.0014.TO.ZS	Demography	electricity	EG.ELC.ACCS.ZS	Technology
female_lfp	SL.TLF.CACT.FE.ZS	Demography	internet	IT.NET.USER.P2	Technology
life_exp	SP.DYN.LE00.IN	Health	mobile	IT.CEL.SETS.P2	Technology

A.2 Reproduction

```
python -m venv .venv && .venv/Scripts/pip install -r requirements.txt
kaggle datasets download -d kaggle/world-development-indicators --unzip -p data/raw
jupyter nbconvert --to notebook --execute --inplace WDI_Analysis.ipynb   # ETL, EDA, figures/
python run_models.py --mode snapshot --seeds 0 1 2 3 4                   # model suite
python run_models.py --mode panel
python -m wdi.story                          # experiments -> results/story, figures/story
python report/make_tables.py                 # LaTeX tables for this report
tectonic report/FINAL_REPORT.tex             # this PDF
python tests/test_models.py && python tests/test_story.py
streamlit run app.py
```

A.3 Repository map

WDI_Analysis.ipynb	ETL, missing-data audit, EDA, trends, comparison, evaluation
wdi/models.py, run_models.py	19-model suite with grouped CV
wdi/story.py	storyline experiments E1.1–E3.3 and report checks
app.py	Streamlit web app
results/, results/story/	model and experiment results (CSV)
figures/, figures/story/	figures used in this report
docs/	storyline, experiment design, task split, app screenshots

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